Lab 5.2: Breaking WEP and WPA and Decrypting the Traffic

Remove Lab Requirement Instructions in your submission

Lab 5.2: Part 1 – Viewing Wireless Networks and Connected Devices

1. Research and describe with the command **aircrack-ng** is for

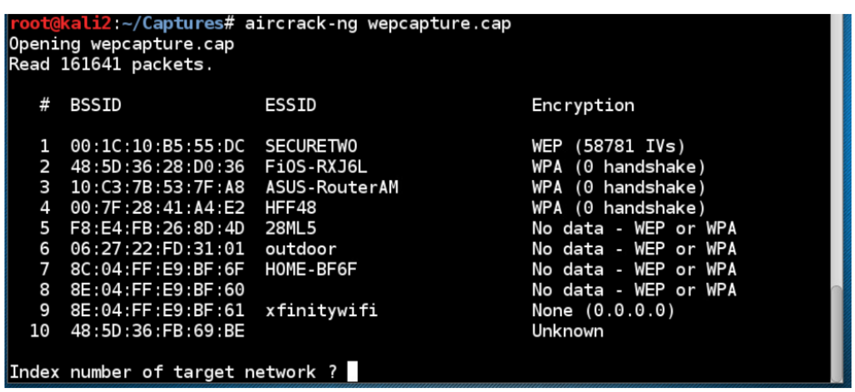
The command is used for auditing Wifi security

2. Research and describe with the command **aircdecap-ng** is for

Used for stripping wireless headers from unencrypted drives

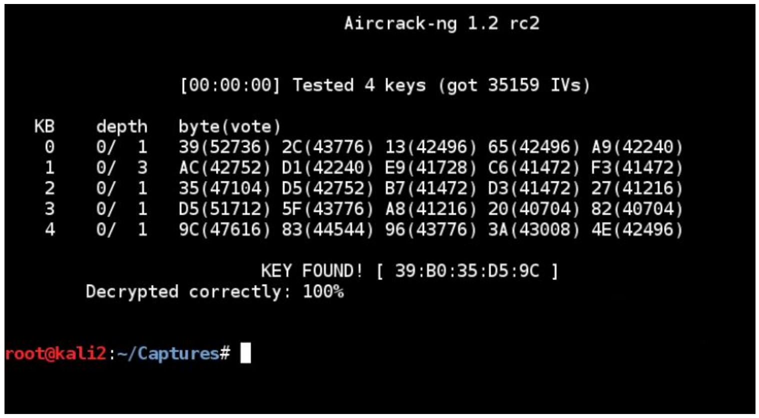
Lab 5.2: Part 2 – Cracking WEP

1. Question 10

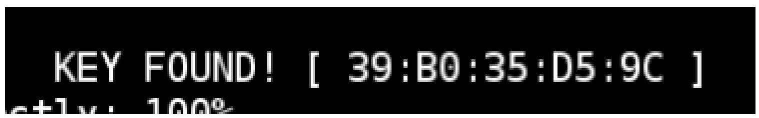


```
root@kali2:~/Captures# aircrack-ng wepcapture.cap
Opening wepcapture.cap
Read 161641 packets.

   #  BSSID              ESSID                     Encryption

   1  00:1C:10:B5:55:DC  SECURETWO                 WEP (58781 IVs)
   2  48:5D:36:28:D0:36  FiOS-RXJ6L                WPA (0 handshake)
   3  10:C3:7B:53:7F:A8  ASUS-RouterAM             WPA (0 handshake)
   4  00:7F:28:41:A4:E2  HFF48                     WPA (0 handshake)
   5  F8:E4:FB:26:8D:4D  28ML5                     No data - WEP or WPA
   6  06:27:22:FD:31:01  outdoor                   No data - WEP or WPA
   7  8C:04:FF:E9:BF:6F  HOME-BF6F                 No data - WEP or WPA
   8  8E:04:FF:E9:BF:60                            No data - WEP or WPA
   9  8E:04:FF:E9:BF:61  xfinitywifi               None (0.0.0.0)
  10  48:5D:36:FB:69:BE                            Unknown

Index number of target network ?
```

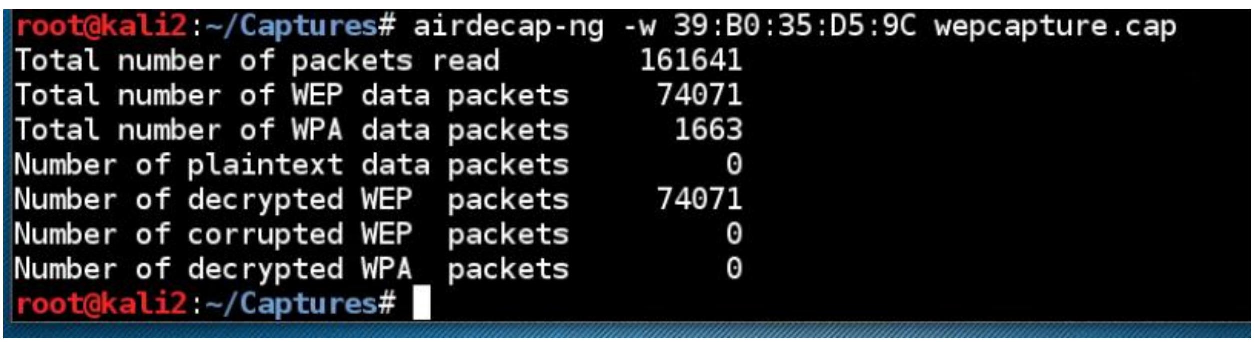
2. Question 11



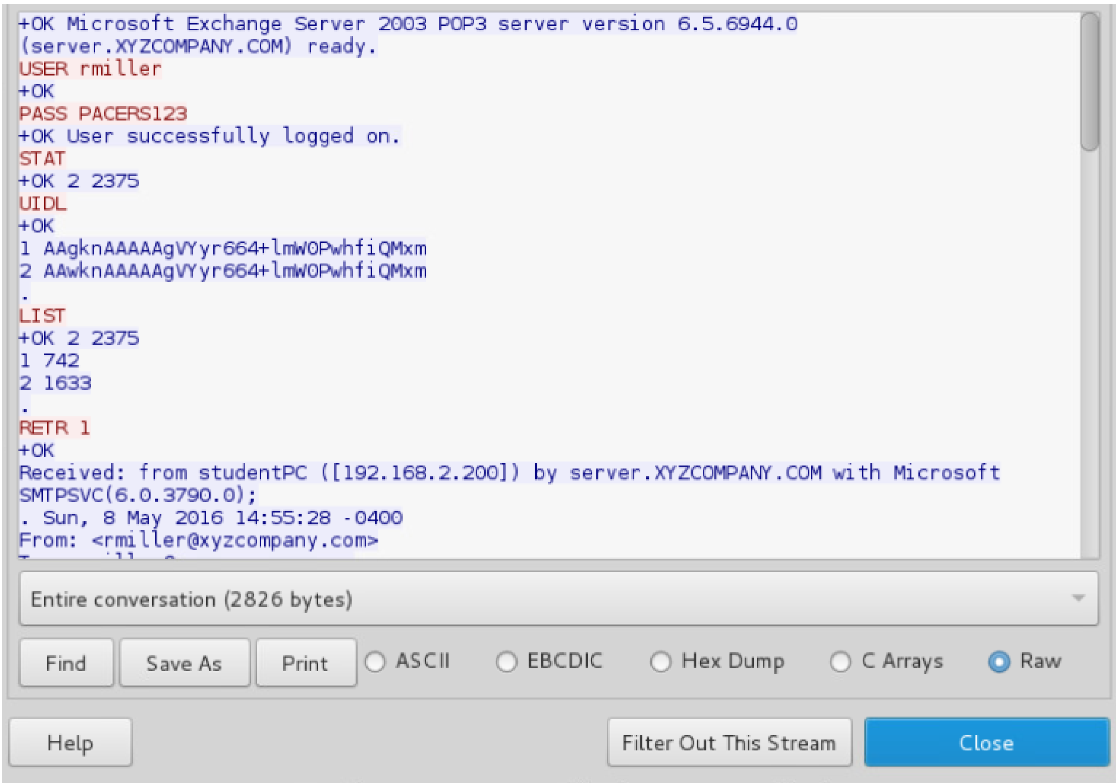
3. Question 12



4. Question 13



5. Question 24



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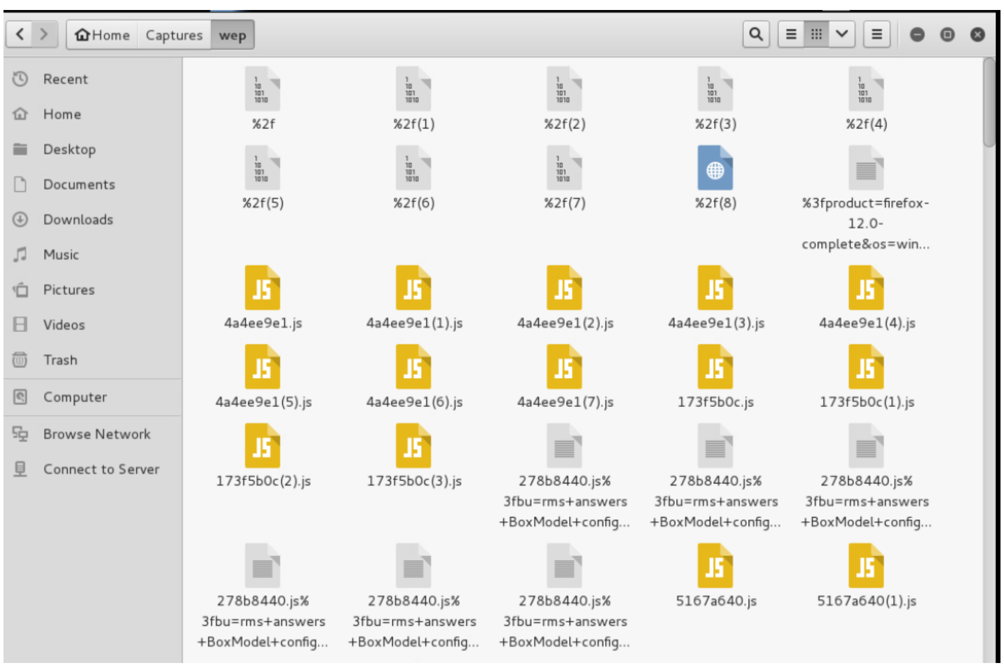
Lab 5.2: Part 3 – Cracking WPA



1. Q
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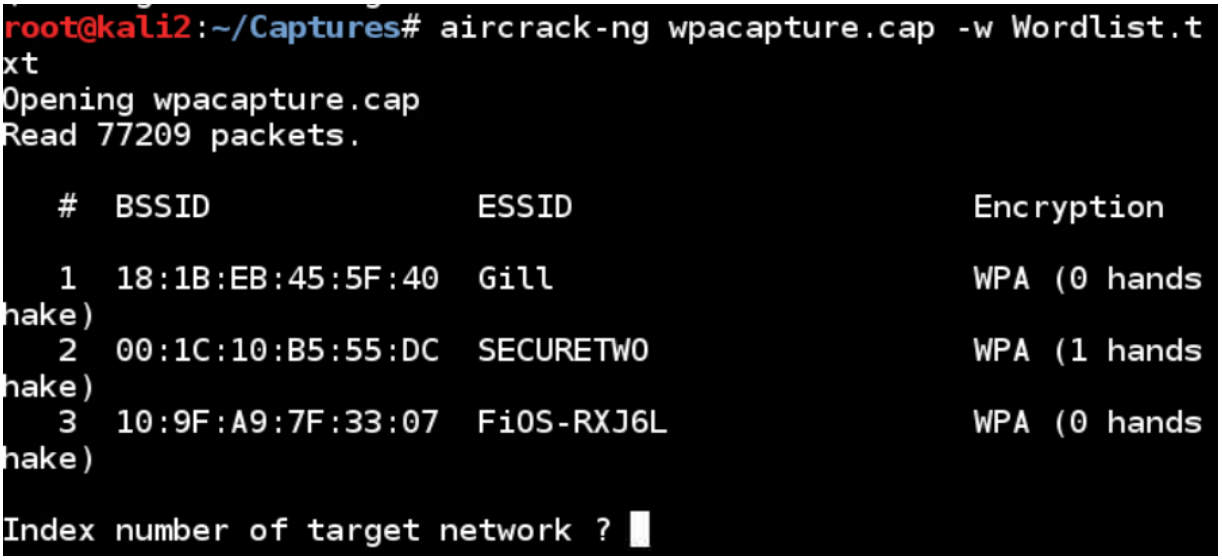
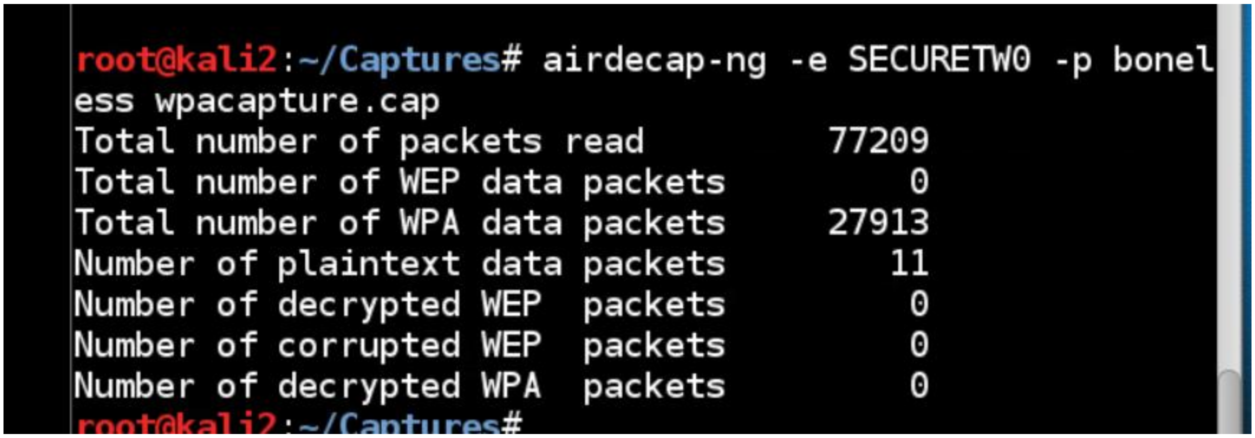


2. Question 7

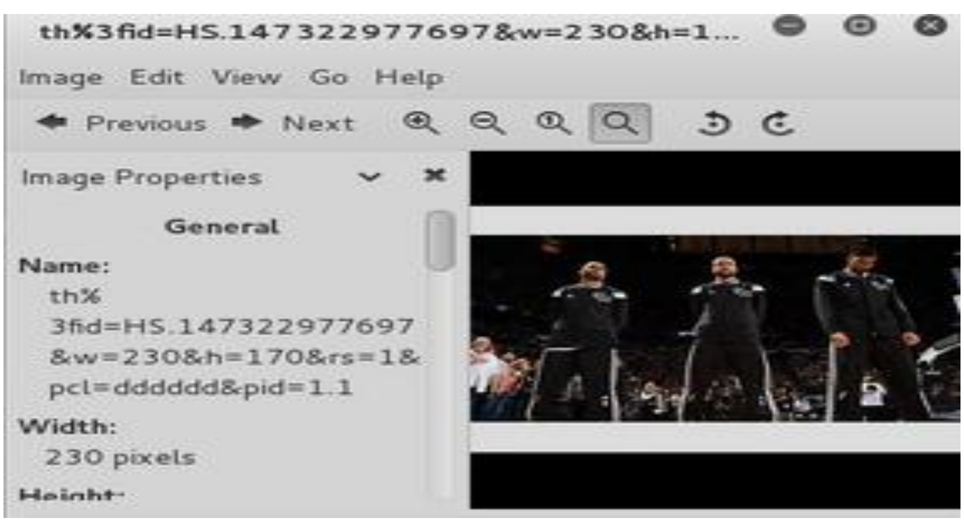
3. Question 8

4. Question 9



5. Question 17

6. Question 25



Lab 5.2: Part 4 – Lessons Learned

Remove Lessons Learned instruction in your submission

Through the use of aircrack-ng and airdecap-ng, I learned how attackers can audit Wi-Fi security and decrypt captured traffic once encryption keys are obtained. The lab highlighted the weaknesses of WEP encryption, particularly its predictable key scheduling and IV reuse, making it highly susceptible to attacks. In contrast, WPA offered improved security through dynamic key exchange and stronger encryption, though it remains vulnerable to dictionary attacks if weak passwords are used. Capturing WPA handshakes and using wordlists to crack passwords emphasized the importance of strong credentials in network defense.